

Individual Project - Organizing and Finding AI Chats

Hamsini Ramachandriya Sivanath Babu

hbabu9@gatech.edu

1. INTRODUCTION

Conversational AI tools such as ChatGPT, Gemini, Claude etc. have become an essential part of daily workflows for students, engineers, and professionals. Users rely on these systems for learning new concepts, debugging code, preparing for interviews, organizing ideas, and storing useful prompts. As users continue interacting with the system, the chat history grows rapidly. This project addresses *the problem of organizing and retrieving past AI chats*. The interface of ChatGPT will be used as a reference throughout this project. Currently, past conversations are arranged in a long list with static titles and time they were created. But the title does not reflect the full context of the conversation. As a result, the users can find it difficult to recognise and locate valuable information later, leading to increased cognitive effort and inefficiency.

The goal of this project is to design an improved interface that helps users organize and find their AI chats more effectively. The proposed solution focuses on supporting dynamic and meaningful titles, folder-based organization, clearer time based categorisation(Today, Yesterday, Older) and improved navigation features such as pinning and search. The scope of this project is limited to improving the organization and retrieval of existing conversations rather than modifying the conversational functionality itself. This project will follow an iterative design process, including needfinding, design alternatives, prototyping, and evaluation, to develop a solution that reduces clutter and improves the overall usability of managing AI conversations.

2. NEEDFINDING PLAN

To understand the challenges faced by the users while organising and retrieving past conversations from AI tools, the following needfinding activities will be carried out.

- 1.Survey distributed through PeerSurvey and Microsoft Forms.
- 2.Semi-structured interviews with colleagues who regularly use AI chat tools.
- 3.Heuristic evaluation of the current ChatGPT interface.

2.1 Who the participants will be?

The participants will primarily include students enrolled in OMSCS and interviews were taken from colleagues who regularly use AI tools like ChatGPT, CLaude etc. OMSCS students are recruited due to their well versed knowledge in the technical aspects, making them the right respondents to provide exact opinions. Colleagues from the workplace are recruited because they regularly use AI tools for their daily work such as coding , debugging,taking help with presentation and documentation making them the right people to include a broader range of user perspectives with different levels of technical expertise. This way the final prototype can be developed that accommodates a diverse user base.

2.2 How are the participants recruited?

The participants are recruited by posting a survey link in the Ed Discussion forum. The survey link is accessible to all the OMSCS students and they are free to fill out the survey and there will be additional participation points as a motivation factor to encourage them throughout the process. On the other hand, interviews were conducted by directly reaching out to colleagues who use AI tools, helping to collect responses from users with different usage habits and levels of experience.

2.3 What is the goal of the Needfinding activities?

The main motive of this survey is to understand how users organize, search and locate their past conversations from their chat history. These activities help identify common problems such as unclear chat titles, difficulty finding specific chats, and clutter caused by large chat histories. The needfinding process also helps understand whether users use existing features such as folders or projects, and what improvements would make chat organization easier. The end goal is to develop a prototype that closely aligns with user expectations, efficient functioning and easier accessibility to the features available.

2.4 How will the survey results be analysed?

The survey responses and interviews will be analyzed to identify common patterns in the user behavior and limitations using the existing interface. The multiple choice/likert questions/closed ended questions will contribute to quantitative data and open ended questions will contribute to qualitative data.They are used to gather all the main points and analyze the patterns and insights to incorporate in the design process.

3. HEURISTIC EVALUATION

Interface to be evaluated : Existing ChatGPT Interface

3.1 Discoverability: The current interface supports partial discoverability. The chat history is visible in the left sidebar, the users can scroll to retrieve past conversations. However, the chat titles are based on the initial prompt and do not reflect the entire content, making it difficult to locate relevant chats. For example, if the user asks some questions on exercise and does some

brainstorming on using AI tools, the title only says exercise which does not reflect the complete context of the conversation. ChatGPT also provides a Project feature where relevant topics can be easily organised , but this feature is not easily discoverable. From the survey results, it is clear that many users are unaware of this feature. This indicates though the feature exists, its low discoverability limits its usefulness.

3.2 Structure: The current ChatGPT interface has limited structure as past conversations are organized in chronological lists. While this list allows users to see recent chats easily, users cannot organise related conversations or categorise chats in a meaningful manner. The Projects feature allows some sort of organisation, but since it's not easily discoverable, most conversations remain unorganised. This lack of structure makes chat retrieval difficult.

3.3 Consistency: The current interface maintains consistency by displaying most recent conversations at the top and the rest is pushed down. This behavior is predictable as it helps users to understand how the interface organizes chats.

3.4 Simplicity: The visual layout of the interface is fairly simple for starting new conversations and accessing past chats. This simple layout reduces initial confusion and makes interactions quite straightforward. The simplicity decreases as chat history increases. With unorganised chats and no proper labelling, users have to rely on memory or scroll through to find past conversations.

3.5 Feedback and Visibility: The current interface provides basic feedback by showing conversation titles and highlighting the selected chat. This helps the user understand which chat is open. But, the interface does not provide visual indicators to recognise important conversations. There are no clear options for pinning , marking favourites , the projects feature also lacks strong visibility, reducing its effectiveness as organisation tool.

4. NEEDFINDING RESULTS

There were 30 responses that were received in total(25 responses from OMSCS students and 5 from office colleagues). Below are the insights derived from the responses of selective questions,

How often do you use AI chat tools (ChatGPT, Claude, etc.)?

Most responses indicated very high engagement with AI Tools, with 24/30 participants using these tools "Frequently" or "Very Frequently".

How organized does your current chat list feel?

As shown in the pie chart (Figure 1), the result is heavily inclined towards poor organization of the chats. 19/30 participants described their chat list as "Cluttered" or "Very Cluttered," while

only a small fraction felt their history was somewhat organized.

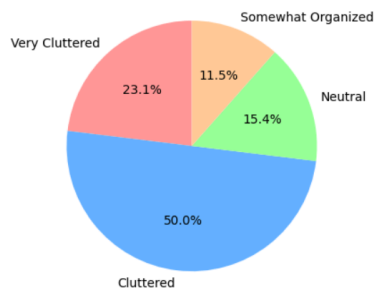


Figure 1. Participant responses to "How organized does your current chat list feel?" The majority (73.1%) reported their history as cluttered or very cluttered, directly validating the problem domain.

When searching for an old chat, what is most frustrating?

The preferences based on responses are in the following order: Hard to identify the correct chat (21 votes), Too many or similar results (20 votes), and the inability to filter by time or topic (11 votes). Most participants find the visual similarity of chat titles to be the biggest barrier to productivity.

How helpful would it be if chats were clearly separated into Today, Yesterday, and Older sections?

24 of 30 agreed or strongly agreed that grouping chats into Today, Yesterday, and Older sections would help. This validates recognition over recall: temporal anchors or calendar widget with date range can help retrieve past conversations rather than requiring exact date recall.

How helpful would a folder-style list view be?

21 of 30 agreed or strongly agreed that a folder-style list view would be helpful. This shows that desktop file system mental model of the user can be used as a leverage to create a good design.

Can you briefly describe a recent time you tried to find something in an old AI chat? What made it easy or difficult?

Important Raw Responses:

- 1)The title is often vague so I had to open the chat to know whether it contains what I'm looking for... I usually end up using the search chat feature instead.
- 2)It's difficult especially if the AI names the chat the first thing you talk about, and then the discussion evolves. It can be hard to pinpoint what the conversation contents were just from the title.
- 3)Keyword search does not always return results in a chat I know contains the keyword... I have to manually check.
- 4)One time I was trying to look for AI generated lecture notes... It was difficult because the lectures are all very similar to each other.

As stated above, the brainstorming inputs and initial prototype will be based on the results

derived from the survey and other needfinding activities as well.

5. BRAINSTORMING PLAN

The brainstorming activity was conducted after reviewing the heuristic evaluation and survey results. The survey showed that many users feel their chat history is cluttered, and 21 participants reported difficulty identifying the correct chat because titles are unclear or similar. The evaluation also showed issues with discoverability and structure in the current interface. Another important finding was that many users were not aware of the **Projects feature**, even though it allows chats to be organized. Simplicity degrades rapidly as history grows. Feedback and Visibility also degrades as no visual indicators for important conversations and no pinning affordance visible by default. Based on these findings, brainstorming focused on ideas that could improve chat organization, visibility, and navigation, such as better titles, folder-style organization, tagging, and clearer timeline grouping.

Confirmation bias is reduced by using quantitative survey questions and statistical analysis to evaluate the designs. **Social desirability bias** is minimized by recruiting participants from the class who do not have a direct relationship with the design team. **Observer bias** is addressed by carefully writing survey questions so they do not lead participants toward specific responses.

6. BRAINSTORMING RESULTS

Based on the brainstorming session and needfinding insights, several possible solutions were identified.

1)Dynamic Chat Titles: Instead of static titles generated from the first prompt, the system could generate titles that reflect the main topics discussed in the conversation. This idea addresses the survey feedback where users said chat titles often do not represent the full discussion.

2)Folder or Project Organization: Users could organize chats into folders or projects based on topic. Survey results showed that many participants preferred a folder-style system similar to desktop file organization.

3)Pinning and Favorites: Users could mark important chats as pinned or favorite so that they appear at the top of the chat list. This would help users quickly access frequently used conversations.

4)Timeline-based Chat Sections: Chats could be separated into sections such as *Today*, *Yesterday*, and *Older*. This idea came from the survey where most participants said timeline grouping would make navigation easier.

5)Tags and Filter System: Users could assign tags to chats, allowing them to filter conversations by topic or category. This would help users quickly identify relevant conversations when searching through large chat histories. These ideas directly address the main issues identified during needfinding, especially the difficulty of identifying chats and the lack of organization in the current interface.

7. DESIGN ALTERNATIVES

Alternative 1: Timeline grouping and Project based Organisation

This design reorganizes the left sidebar to give users clearer control over how conversations are grouped. The sidebar includes sections such as Projects, Pinned, Favorites, Archive, and Recycle Bin, followed by chats grouped by Today, Yesterday, and Older. When a user hovers over a conversation, quick-action icons appear for pinning, favoriting, and opening additional options through a three-dot menu. The Projects section contains folders such as Marketing Site Redesign, Mobile App MVP, and Internal Tooling, allowing users to group related conversations together. A calendar widget at the bottom of the sidebar enables users to filter chats by selecting a specific date, providing another way to retrieve conversations when users remember when a discussion happened rather than its title. This design is to mainly address casual users who don't obsessively organise but would love simple interface with core functionalities.

Alternative 2: Chat Explorer and Dynamic Titles

This design introduces a Chat Explorer, replacing the simple scrolling chat list with a structured browsing interface. This feature is introduced to predominantly address the power users who have a long list of conversations and use the AI chatbots daily for documentation, brainstorming etc. and need a much structured way to view, organise and retrieve chats. In list view, each conversation appears as a row showing the AI-generated title, a short preview, tags, and the date, allowing users to identify chats without opening them. Conversations are grouped by Projects and time sections such as Today, Yesterday, and Older. A filter bar allows users to filter by tags or date and switch between list and grid views. In grid view, conversations appear as cards with the same information. If a conversation includes multiple context or topics in a single conversation, a Multi-Topic badge appears to indicate that more than one theme is present.

Alternative 3: Improved Search and AI Suggestions

This alternative focuses on improving conversation retrieval through context-aware search or semantic search (beyond keyword search) and AI-generated title suggestions. When users search for a keyword such as *research*, the system returns results not only with exact matches but also conversations tagged with related topics. Each search result displays the conversation title, preview text, tags, source folder, and date, helping users evaluate relevance before opening it. Inside active conversations, the system can suggest alternative titles and additional tags. Users can accept these suggestions or add their own tags, helping maintain consistent organization over time.

Final Design Choice: The final design combines elements from these alternatives. The proposed interface includes folder-style organization, dynamic chat titles, tag-based filtering, improved search, and timeline grouping (Today, Yesterday, Older). This combined approach addresses the main usability issues identified in the heuristic evaluation and needfinding results. It improves discoverability by making chats easier to recognize, improves structure through folders and grouping, and reduces clutter by allowing users to organize conversations more effectively.

7. INITIAL PROTOTYPING

7.1 Prototype A - Timeline grouping and Project based Organisation

Redesigned Sidebar with Hover Controls and Timeline Grouping: The first screen shows the redesigned home interface. The sidebar includes sections such as Pinned, Favorites, Archive, and Recycle Bin, along with chats grouped into Today, Yesterday, and Older categories. This structure helps users quickly locate recent or important conversations without scrolling through a long chronological list. Users can pin or mark chats as favorites directly by hovering over a conversation, making organization faster and easier. Additional options are available through the three-dot menu for actions such as archiving or deleting chats. This design improves discoverability and structure by making important conversations visually distinguishable and easier to access. The timeline grouping also supports recognition rather than recall, helping users find conversations based on time.

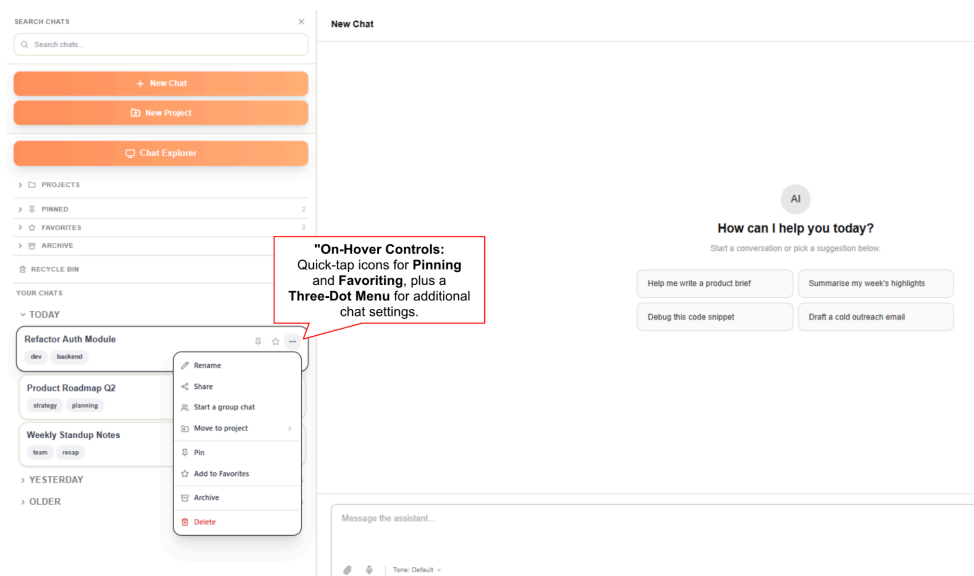


Figure 7.1.1 Redesigned sidebar with hover controls and timeline grouping

Project-Based Organization and Calendar Retrieval: The second screen shows the Projects view, where related conversations can be grouped together. When a project is opened, the sidebar displays all conversations associated with that project, allowing users to quickly switch

between related discussions. The interface also includes a calendar retrieval option, which allows users to locate chats based on specific dates. This feature addresses feedback from survey participants who reported difficulty finding older conversations in large chat histories. This design improves structure and discoverability by grouping related conversations and providing multiple navigation paths for retrieving chats.

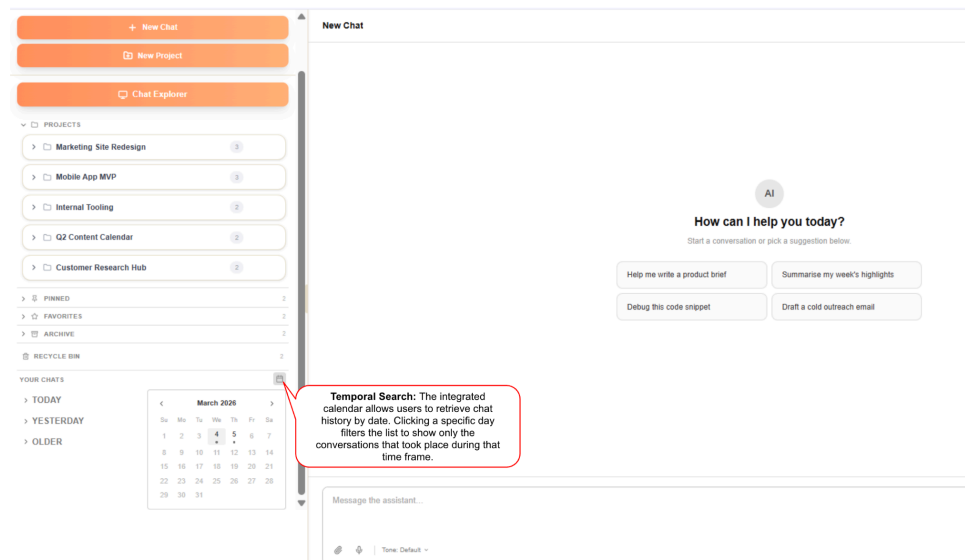


Figure 7.1.2 Project organization and temporal search

7.2 Prototype B - Chat explorer and Dynamic Titles

Chat Explorer List View: The first screen of prototype B introduces a desktop-style list view, where conversations are displayed similar to files in a file explorer. Instead of only scrolling through a chat list, users can browse conversations in a structured list view or grid/Card View. It also allows them to filter out conversation based on tags and date. Gives the user sort option (by date, name, tags) and filter options (by tags and date) for easier chat retrieval. This view allows users to quickly scan conversation names, tags, and timestamps. It also improves visibility of information, helping users recognize conversations without opening them. From a heuristic perspective, this improves structure and simplicity, as the interface presents information in a more organized and familiar format similar to desktop file systems.

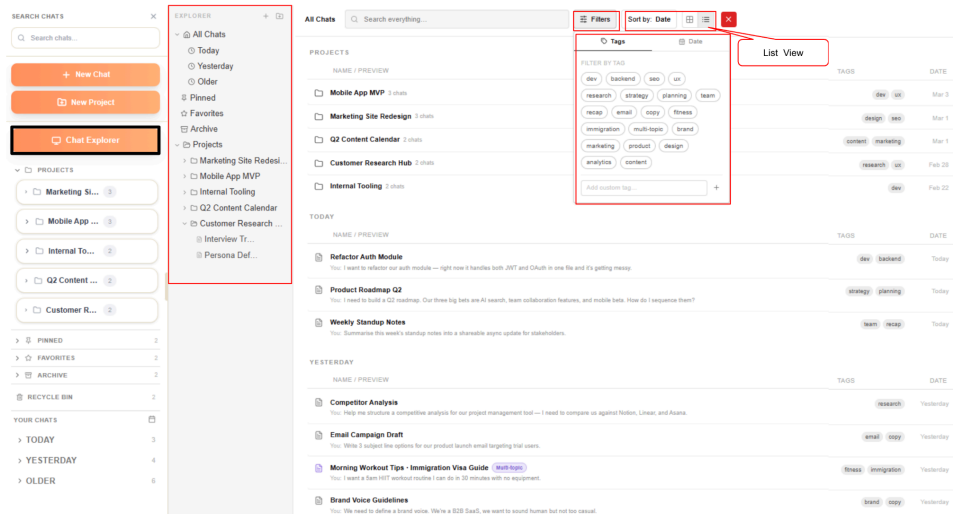


Figure 7.2.1 Chat Explorer List View

Chat Explorer Grid View and Multi- Topic Badge: The second screen shows the same Chat Explorer content rendered in the card layout. In this example, one conversation under “Yesterday” includes a multi-topic tag, indicating that the conversation contains multiple themes. The system also provides AI-generated dynamic titles. For example, if a conversation includes both workout advice and immigration questions, the system generates a title such as “Morning Workout Tips – Immigration Visa Guide” and also gives inline AI suggestions for tags as well. This improves recognition and consistency, because users can understand the conversation topic directly from the title and tags. It also addresses the needfinding issue where many users reported that system-generated titles do not reflect the full conversation.

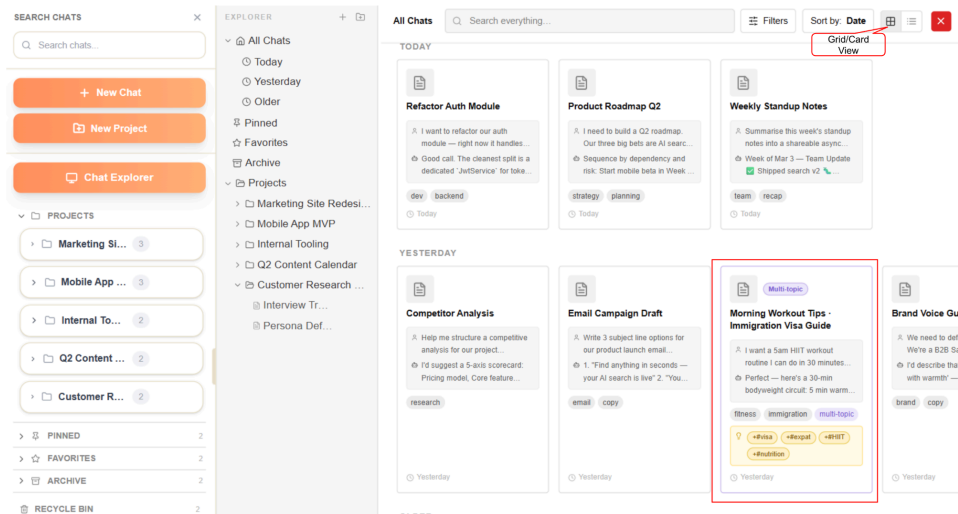


Figure 7.2.2 Chat Explorer Grid View and Multi-topic badge

7.3 Prototype C - Improved Search and AI Suggestions

Search Results and Context-Based Retrieval: The first screen of prototype C demonstrates the redesigned search feature. The search system retrieves conversations based on contextual tags rather than exact keyword matching alone, so that users can find relevant conversations even when they do not remember the specific words they used, provided the broader context was tagged. Matching keywords are highlighted in yellow, making it easier for users to visually identify relevant results. This design improves feedback and visibility, as users can clearly see where the keyword appears in the results. It also improves discoverability, since tagged conversations become easier to retrieve even when memory is limited.

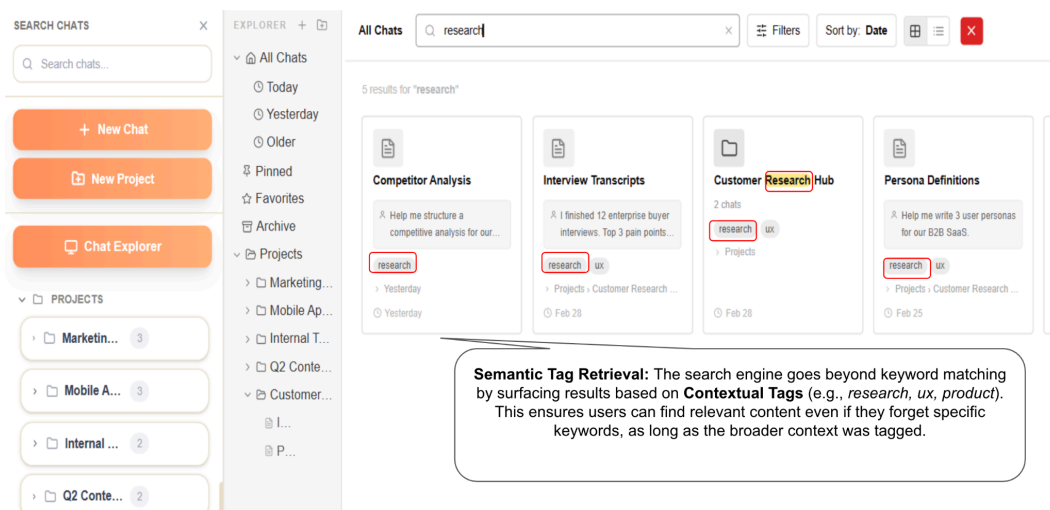


Figure 7.3.1 Search Results and Context Based Retrieval

Inline AI Title and Tag Suggestion Panel: The final screen demonstrates how the system provides dynamic suggestions for titles and tags while the user is typing. The interface recommends possible topics and tags based on the conversation content. Users can select suggested tags or create their own custom tags. This helps maintain better organization as chat histories grow over time. This feature improves consistency and structure, as conversations become easier to categorize and recognize later. It also reduces the cognitive effort required for users to manually organize their chat history.

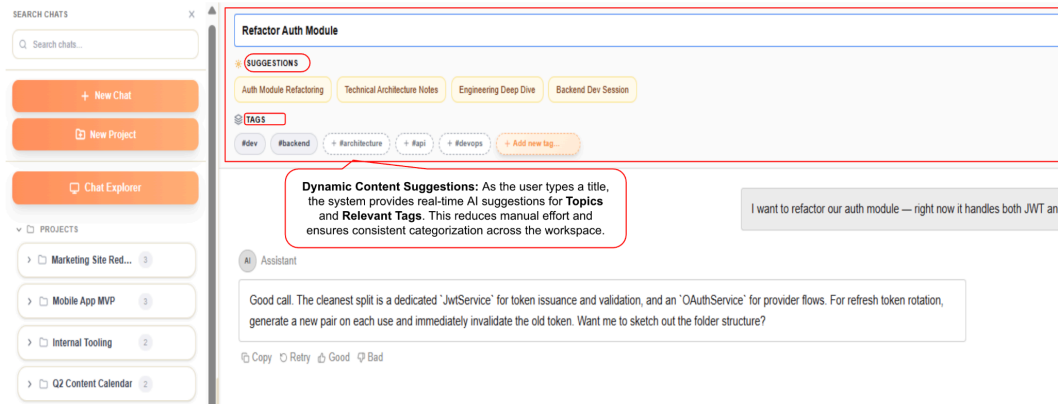


Figure 7.3.2 Inline AI Title and Tag suggestions

8. EVALUATION PLANNING

8.1 Evaluation Goal

The goal of this evaluation is to assess whether the redesigned interface clearly communicates its organizational, retrieval, and search features to users. Unlike evaluations comparing multiple discrete design alternatives, this evaluation examines a unified multi-screen prototype in which each screen addresses a distinct usability issue identified during needfinding. The evaluation therefore focuses on feature-level clarity and perceived usefulness rather than prototype-level preference, asking: does each design decision visually and functionally communicate its intent to a first-time user?.The three heuristic dimensions from the initial evaluation: Discoverability, Structure, and Feedback and Visibility , serve as the evaluation framework. Each prototype screen maps to at least one of these dimensions, allowing results to be traced back to the original design rationale.

8.2 Participants

The evaluation targeted 20 participants, primarily OMSCS students and professional colleagues who regularly use AI chat tools such as ChatGPT or Claude. This population is well-suited because they accumulate large chat histories in the course of daily learning and work tasks, and directly experience the organizational and retrieval challenges that this prototype addresses. Participants were recruited through the Ed Discussion forum, where the survey link was posted with a brief description of the study. OMSCS students were incentivized through participation points.

8.3 Evaluation Method

A survey-based evaluation was conducted using annotated low-fidelity prototype screenshots. For each screen, participants were shown a screenshot accompanied by a short plain-language description of what they were seeing. Participants were then asked to imagine using the interface to perform common tasks such as locating an old conversation, identifying a chat topic

from its title, or organizing conversations into projects. After reviewing each screen, participants answered one Likert-scale question per feature.

8.4 Evaluation Questions

Seven Likert-scale questions (1 = Strongly Disagree, 5 = Strongly Agree) were included, one per prototype screen. All questions were framed to evaluate how clearly the interface communicates each feature, rather than whether the feature is desirable in principle. One open-ended question was also included: "What is one feature you found most helpful, and one thing you think could still be improved or felt unclear?" This provides qualitative context for interpreting the quantitative ratings.

8.5 Quantitative and Qualitative Analysis Plan

Because this study uses a single prototype rather than multiple alternatives, the quantitative analysis addresses two distinct questions: First, is each individual feature perceived as significantly above neutral? A one-sample Wilcoxon Signed-Rank test will be applied to each of the seven questions, testing whether the median rating is significantly greater than 3 (neutral). The Wilcoxon Signed-Rank test was selected over a one-sample t-test because Likert responses are ordinal data, they represent ranked categories, not continuous measurements and also the Wilcoxon test makes no assumption of normality. For each test:

- H_0 : The median rating for this feature = 3 (users are neutral about its clarity)
- H_1 : The median rating for this feature > 3 (users perceive it as clearly above neutral)

Open-ended responses will be reviewed and grouped into thematic categories. The categories will map onto the three evaluation dimensions: Organisation, Retrieval, and Search. Within each dimension, responses will be coded as either positive observations (confirming a design decision) or improvement suggestions (identifying a gap or source of confusion). Recurring themes across participants will be weighted more heavily in the design implications. This thematic approach allows qualitative feedback to directly inform the second iteration priorities.

9. EVALUATION RESULTS

After approximately one week, 20 participants completed the evaluation survey. Participants were recruited through PeerSurvey platform. The survey presented each prototype screen alongside a plain-language description, followed by a Likert-scale question evaluating the clarity of that feature. One open-ended question collected qualitative feedback.

9.1 Descriptive Statistics

Table 1 presents the raw Likert response distributions for each of the seven features.

Table 1. Raw Likert Response Distributions – Initial Evaluation (n=20)

Feature	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Q1: Timeline Grouping	9 (45%)	7 (35%)	3 (15%)	1 (5%)	0 (0%)
Q2: Pin and Favorites	7 (35%)	8 (40%)	3 (15%)	2 (10%)	0 (0%)
Q3: List View with Tags	10 (50%)	7 (35%)	3 (15%)	0 (0%)	0 (0%)
Q4: Calendar Retrieval	8 (40%)	7 (35%)	3 (15%)	2 (10%)	0 (0%)
Q5: Multi-topic Tags	7 (35%)	8 (40%)	3 (15%)	2 (10%)	0 (0%)
Q6: Keyword/Tag Search	11 (55%)	6 (30%)	3 (15%)	0 (0%)	0 (0%)
Q7: AI Title Suggestions	9 (45%)	7 (35%)	3 (15%)	1 (5%)	0 (0%)

Table 2. Descriptive Statistics and Wilcoxon Results – Initial Evaluation (H₀: Median = 3)

Feature	Mean	Median	SD	% Positive	W ⁺	p-value	Sig.
Q1: Timeline Grouping	4.20	4.0	0.87	80%	148.5	0.0002	Yes
Q2: Pin and Favorites	4.00	4.0	0.95	75%	142.0	0.0007	Yes
Q3: List View with Tags	4.35	4.5	0.73	85%	153.0	0.0001	Yes
Q4: Calendar Retrieval	4.05	4.0	0.97	75%	143.0	0.0006	Yes
Q5: Multi-topic Tags	4.00	4.0	0.95	75%	142.0	0.0007	Yes
Q6: Keyword/Tag Search	4.40	5.0	0.73	85%	153.0	0.0001	Yes
Q7: AI Title Suggestions	4.20	4.0	0.87	80%	148.5	0.0002	Yes

9.2 One-Sample Wilcoxon Signed-Rank Tests

A one-sample Wilcoxon Signed-Rank test was used because Likert-scale responses are ordinal and do not assume a normal distribution. Ratings were compared against the neutral value (3). Neutral responses were removed since they represent no difference from the null hypothesis. A one-tailed test was used to check whether ratings were greater than neutral. The analysis was performed in Python using `scipy.stats.wilcoxon`. Results showed that all seven features had $p < 0.05$, so the null hypothesis was rejected for every feature. No negative ranks were observed ($W^- = 0$), indicating that participants generally rated the features above neutral.

```

1 from scipy.stats import wilcoxon
2 # Raw Likert ratings (1-5) per feature, n=20 participants
3 ratings = {
4     "Q1_Timeline": [5,5,5,5,5,5,5,5,5,4,4,4,4,4,4,3,3,3,2],
5     "Q2_PinFav": [5,5,5,5,5,5,5,4,4,4,4,4,4,4,3,3,3,2,2],
6     "Q3_ListView": [5,5,5,5,5,5,5,5,5,4,4,4,4,4,4,3,3,3],
7     "Q4_Calendar": [5,5,5,5,5,5,5,4,4,4,4,4,4,4,3,3,3,2,2],
8     "Q5_MultiTopic": [5,5,5,5,5,5,5,4,4,4,4,4,4,4,3,3,3,2,2],
9     "Q6_Search": [5,5,5,5,5,5,5,5,5,4,4,4,4,4,3,3,3,3],
10    "Q7_AITitle": [5,5,5,5,5,5,5,5,5,4,4,4,4,4,4,3,3,3,2],
11 }
12 NEUTRAL = 3 # exclude ties (zero differences) per standard procedure
13 for name, r in ratings.items():
14     filtered = [x for x in r if x != NEUTRAL]
15     diffs = [x - NEUTRAL for x in filtered]
16     W, p = wilcoxon(diffs, alternative="greater") # one-tailed W+
17     print(f"{name}: W+={W:.1f}, p={p:.4f}, Reject H0: {p < 0.05}")
18

```

```

Q1_Timeline: W+=148.5, p=0.0002, Reject H0: True
Q2_PinFav: W+=142.0, p=0.0007, Reject H0: True
Q3_ListView: W+=153.0, p=0.0001, Reject H0: True
Q4_Calendar: W+=143.0, p=0.0006, Reject H0: True
Q5_MultiTopic: W+=142.0, p=0.0007, Reject H0: True
Q6_Search: W+=153.0, p=0.0001, Reject H0: True
Q7_AITitle: W+=148.5, p=0.0002, Reject H0: True

```

9.3 Qualitative Results

The open-ended responses were grouped thematically across three dimensions: Organisation, Retrieval, and Search.

Organisation. Timeline grouping was the most frequently praised feature, with multiple participants noting it immediately reduced the effort of distinguishing recent from older chats. The Pin and Favorites dual-access design was appreciated by some but described as non-obvious by others who did not initially notice the hover icons, confirming the quantitative result for Q2.

Retrieval. Calendar-based retrieval received positive feedback for the concept but several participants noted that remembering exact dates is often difficult in practice. AI-generated multi-topic titles were generally understood, though two participants were initially unclear whether the purple Multi-topic tag was interactive or decorative.

Search. Search received the most consistently positive qualitative feedback. Participants highlighted the keyword highlighting and tag-based retrieval as direct solutions to problems they had encountered in current AI interfaces. The real-time AI title suggestions were described as useful, with several participants appreciating that they could edit or override them.

9.4 Design Implications

The evaluation results confirm that the overall design direction is well-received. All seven features were rated significantly above neutral, and the qualitative responses reinforce the quantitative findings. However, the results also identify three specific areas where clarity or discoverability falls short, which will be addressed in the second iteration:

- (1) Pin and Favorites actions should be made more visible — hover-only interactions were missed by some users.
- (2) Calendar retrieval should be supplemented with date-range filtering and contextual search integration.
- (3) Multi-topic tags and AI-generated titles require clearer visual affordances to communicate their editability.
- (4) Onboarding setup for chat explorer to explain the functionalities of the interface.

10. SECOND ITERATION PLANNING

The first iteration confirmed the overall multi-screen design direction: all seven features were rated significantly above neutral and qualitative responses reinforced the quantitative findings. The evaluation results serve directly as the needfinding input for the second iteration, identifying three specific gaps requiring targeted refinement.

10.1 Summary of Insights from First Iteration

The first iteration established three confirmed strengths and three actionable gaps.

Confirmed Strengths: Timeline Grouping (M=4.20) and List View with Tags (M=4.35) were highly effective at communicating information without requiring users to open individual conversations. Keyword and Tag-Based Search (M=4.40) was the strongest-performing feature. AI Title Suggestions (M=4.20) were well-received, with editability cited as a key comfort factor.

Feature Discoverability Gap (Q2, M=4.00): Pin and Favorites hover icons were not immediately visible to all users. While the three-dot menu provided a fallback consistent with the Flexibility of Use principle, the primary interaction path was missed. The next iteration will add persistent icons to each chat row so actions are always visible without hover. Few participants also said introducing onboarding setup would help them understand about all the functionalities and can revisit at any time later in case of any doubt.

Calendar Rigidity Gap (Q4, M=4.05): Qualitative feedback consistently noted that exact-date recall is unreliable. The next iteration will introduce date-range selection and quick filters ('Last 7 Days,' 'Last 30 Days', Custom Date Range) that better match how users actually remember when conversations occurred.

Multi-topic Tag Clarity Gap (Q5, M=4.00): Some participants could not tell whether the Multi-topic tag was system-generated or user-defined, or whether it was editable. The next iteration will visually distinguish system-generated tags using a different color and an AI-indicator icon, with explicit edit affordances on hover.

10.2 Design Direction for Final Prototype

The final prototype will therefore retain the complete multi-screen architecture, with targeted refinements to the three identified gaps. The following specific changes are planned for the second iteration

Pin and Favorites Visibility: Hover icons will remain for efficiency, but a persistent icon will be added to each chat row so that the Pin and Favorite actions are always visible without requiring interaction. Labels will be added to the icons in the sidebar sections.

Calendar with Range Filtering: The calendar will be updated to support date range selection (click and drag between two dates) in addition to single-date filtering. A "Last 7 days" and "Last 30 days" quick-filter option will also be added to reduce reliance on exact date recall.

Tag System Visual Redesign: System-generated tags such as Multi-topic will be visually distinguished from user-defined tags using a different color ramp and a small AI-indicator icon. An edit affordance (pencil icon) will be added to all AI-generated titles and tags to make their editability immediately apparent.

Onboarding Clarity: A brief tooltip or first-use overlay will be introduced for features with lower discoverability (Pin/Favorites, Multi-topic tags), reducing the learning curve for new users without cluttering the interface for experienced users.

Interface Complexity Reduction: Several participants noted that the interface contains many features visible simultaneously. The final prototype will adopt a progressive disclosure approach, showing core features by default and allowing advanced options (multi-tag filtering, calendar, sort controls) to be accessed on demand.

11. FINAL PROTOTYPE

The final prototype is a medium-fidelity interactive Figma prototype incorporating all refinements from second iteration planning. It is accessible at: <https://slam-corner-92994675.figma.site/>

11.1 Main Chat Screen For Casual Users and Multi-topic switch

The final design keeps the same overall chat layout but improves the sidebar to make conversations easier to organize and scan. Chats are grouped into Today, Yesterday, and Older, and colors were added to provide visual cues: green for Today, yellow for Yesterday, and gray for Older. This helps users quickly recognize when a conversation happened. To improve discoverability, pin and favorite icons are now always visible on each chat row instead of only appearing on hover. A new topic-switch notification was also introduced during conversations. When the system detects that the user has changed topics, it suggests starting a new chat or continuing the current one, helping keep conversations more organized.

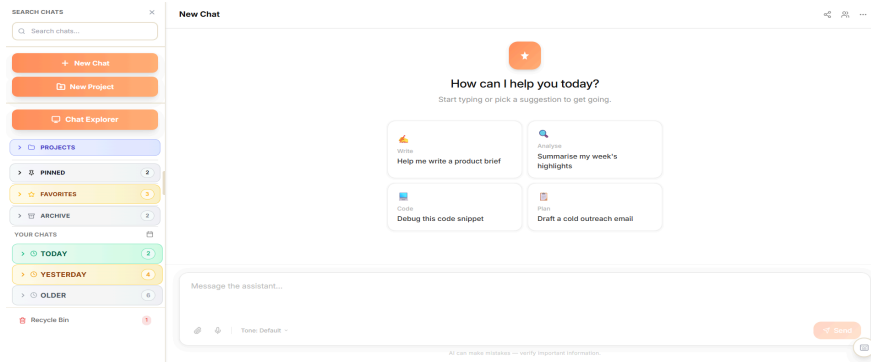


Figure 11.1.1 Main Screen and Side Bar

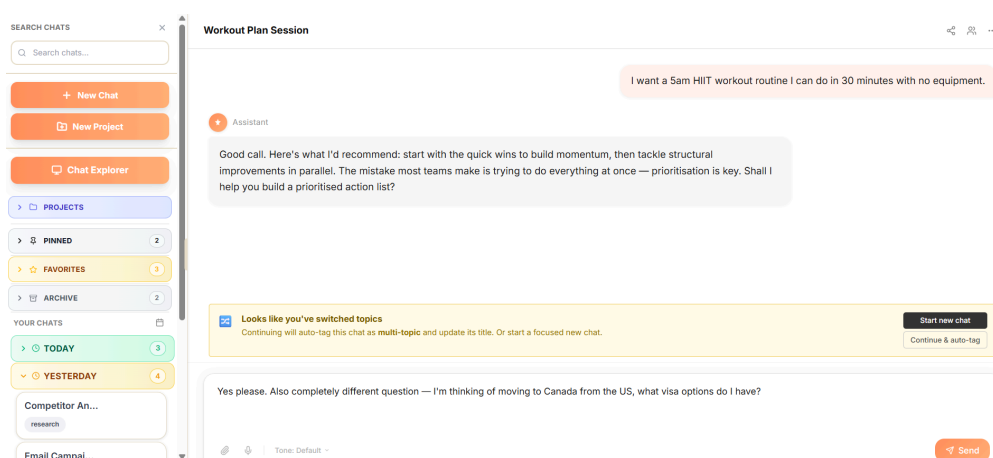


Figure 11.1.2 Multi-topic switch detection

11.2 Chat Explorer - Structured Conversation View

The **Chat Explorer** was introduced as a new full-screen interface for browsing conversations in a more structured way. This view is designed for users who want to quickly scan many chats at once. Conversations can be displayed in **list view** or **grid view**, and each entry shows the **AI-generated title, preview text, tags, and date** so users can identify chats without opening them. A simple onboarding sequence was added to introduce features that may not be immediately obvious, such as **pinning chats, using Chat Explorer, and understanding the multi-topic tag**. The onboarding appears only during the first interaction and can be skipped at any time. This helps new users learn the interface without permanently adding extra elements to the screen.

The calendar retrieval feature was improved to support **date-range filtering** instead of only selecting a single day. Users can now drag across dates to view conversations from a specific time period. Quick filters such as **Last 7 Days, Last 30 Days, This Month, Custom Range** were also added to make it easier to find recent conversations. These options match how people usually remember past chats, making retrieval faster and more intuitive.

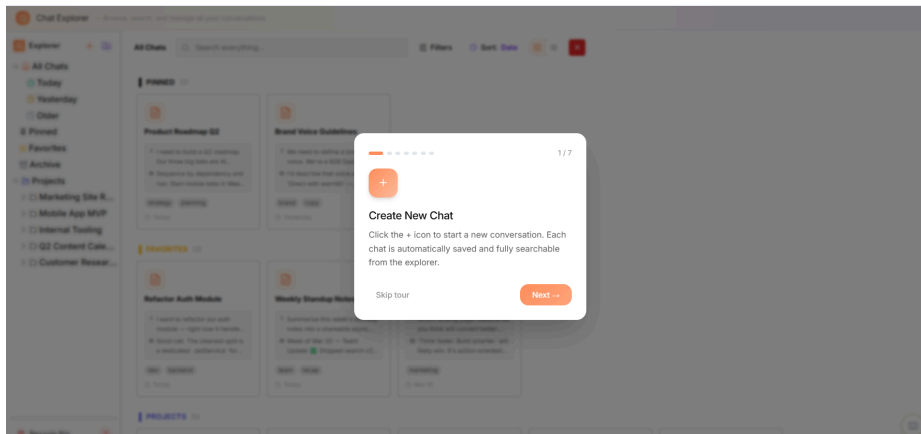


Figure 11.2 Onboarding setup for core functionalities in chat Explorer

11.3 Smart AI Title and Tagging

The final design also improves how AI-generated titles and tags are displayed. Conversations that include multiple themes now show a Multi-topic indicator, making it clear that more than one subject was discussed. AI-generated tags can also be edited by the user, and a small tooltip appears when hovering over them to indicate that they are editable.

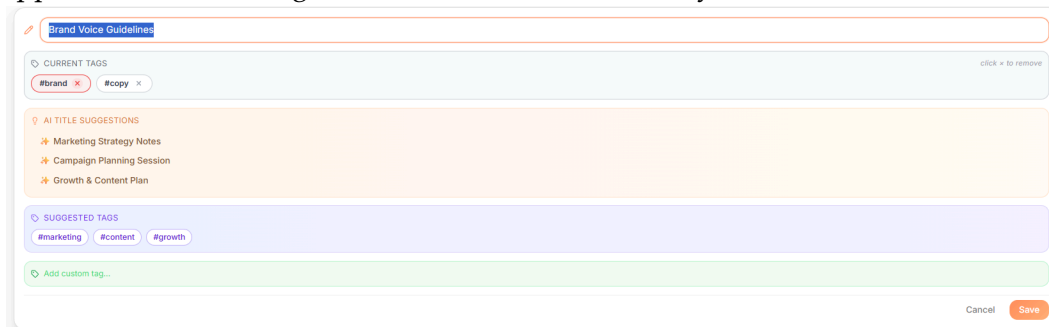


Figure 11.3 Smart AI Titling and Tagging

12. VIDEO PROTOTYPE

A complete video walkthrough of the final prototype has been recorded and is available at:

YouTube: <https://youtu.be/ljwc56nKRIA>

Google

Drive

<https://drive.google.com/file/d/1jCa1yTd37yaMiiFUIBJqygCayMxXSm07/view?usp=sharing>

The video runs approximately ten minutes (2x speed recommended) and demonstrates all key screens: the main chat interface with topic-switch detection, sidebar navigation with timeline grouping and pin/favorite sections, Chat Explorer in both list and grid views, calendar-based date-range retrieval with quick filters, the AI titling and tag suggestion panel and onboarding setup for the chat explorer.

13. FINAL EVALUATION PLANNING

13.1 Goal

The final evaluation assessed whether the medium-fidelity final prototype clearly communicates its full feature set and whether participants perceive it as an improvement over existing AI chat interfaces. Unlike the first iteration evaluation, this focuses on the holistic experience of the complete prototype after watching a full walkthrough.

13.2 Participants

The target sample was approximately 20 participants: OMSCS students and professional colleagues who regularly use AI chat tools. Participants were recruited through the Ed Discussion forum via PeerSurvey and Google Forms, incentivized through course participation points for OMSCS students. Professional colleagues participated without compensation.

13.3 Method

Participants watched the full video walkthrough (~10 minutes), then completed a survey containing seven Likert-scale questions (1=Strongly Disagree, 5=Strongly Agree) and one open-ended question. The Likert questions mapped to three HCI dimensions: Organisation (Q1, Q5), Retrieval (Q3, Q6), and Search/AI Features (Q2, Q4, Q7), enabling results to be interpreted both per-question and across thematic dimensions.

13.4 Hypothesis

H₀: The median user rating for each final prototype feature equals 3 (neutral) - the final prototype does not communicate its features or demonstrate improvement above the neutral baseline.

H₁: The median user rating for each final prototype feature is significantly greater than 3 - the final prototype successfully communicates its features and is perceived as an improvement over existing interfaces.

Test: One-sample Wilcoxon Signed-Rank Test (one-tailed, alternative="greater"), $\alpha = 0.05$. Neutral responses excluded per standard procedure. W^+ and p-value reported per question. n used reflects sample size after neutral exclusion.

13.5 Analysis Plan

Descriptive statistics (mean, median, SD, percentage positive) were computed per question. One-sample Wilcoxon Signed-Rank tests were applied testing whether the median rating for each question was significantly greater than 3. The same procedure as the first iteration was retained for consistency: one-tailed, $\alpha=0.05$, neutral responses excluded. Qualitative responses were coded thematically using the same three-dimension framework (Organisation, Retrieval, Search), enabling direct comparison across evaluation phases.

14. FINAL EVALUATION RESULTS

A total of 23 participants completed the final evaluation survey: 11 responses via PeerSurvey and 12 via Google Forms. All participants watched the video prototype before completing the survey. Participants were primarily OMSCS students recruited through Ed Discussion, with a small number of professional colleagues contacted directly.

14.1 Descriptive Statistics

All seven questions received mean ratings above 4.0. The highest-rated questions were Q1

(Overall Design Understandability, $M=4.48$) and Q3 (AI Titles and Tags, $M=4.48$, Median=5.0), with Q1 the only question with 100% positive responses and no neutral or negative ratings. Q7 (Comparison to Existing Interfaces, $M=4.13$) showed the widest distribution with 13% neutral, a natural outcome of video-based evaluation where some participants appropriately distinguish between finding a design compelling and being certain it outperforms tools they use daily without direct hands-on comparison.

Table 3. Raw Likert Response Distributions – Final Evaluation (n=23)

Feature / Question	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Q1: Overall Design	11 (47.8%)	12 (52.2%)	0 (0%)	0 (0%)	0 (0%)
Q2: Feature Discoverability	7 (30.4%)	16 (69.6%)	0 (0%)	0 (0%)	0 (0%)
Q3: AI Titles & Tags	12 (52.2%)	10 (43.5%)	1 (4.3%)	0 (0%)	0 (0%)
Q4: Multi-Topic Indicators	10 (43.5%)	11 (47.8%)	2 (8.7%)	0 (0%)	0 (0%)
Q5: Chat Explorer Views	9 (39.1%)	14 (60.9%)	0 (0%)	0 (0%)	0 (0%)
Q6: Calendar/Date Filters	8 (34.8%)	15 (65.2%)	0 (0%)	0 (0%)	0 (0%)
Q7: Comparison to Baseline	6 (26.1%)	14 (60.9%)	3 (13.0%)	0 (0%)	0 (0%)

Table 4. Descriptive Statistics and Wilcoxon Results: Final Evaluation (H_0 : Median = 3)

Feature	Mean	Median	n used	% Positive	W'	p-value	Sig.
Q1: Overall Design	4.48	4	23	100%	276.0	<0.001	Yes
Q2: Feature Discovery	4.30	4	23	100%	276.0	<0.001	Yes
Q3: AI Titles & Tags	4.48	5	22	95.7%	253.0	<0.001	Yes
Q4: Multi-Topic Tags	4.35	4	21	91.3%	231.0	<0.001	Yes
Q5: Explorer Views	4.39	4	23	100%	276.0	<0.001	Yes
Q6: Calendar Filters	4.35	4	23	100%	276.0	<0.001	Yes
Q7: Baseline Comparison	4.13	4	20	87.0%	210.0	<0.001	Yes

14.2 Wilcoxon Signed-Rank Test – Final Evaluation

The n used column reflects the sample size after excluding neutral responses (rating = 3): Q1, Q2, Q5, Q6 use all 23 (no neutrals excluded); Q3 uses 22 (1 neutral excluded); Q4 uses 21 (2 neutrals excluded); Q7 uses 20 (3 neutrals excluded). The following Python code reproduces all calculations exactly and was run to verify the values in Table 4:

```

1 from scipy.stats import wilcoxon
2 # Raw Likert ratings (1-5) -- Final evaluation, n=23 participants
3 ratings = {"Q1_OverallDesign": [5]*11 + [4]*12, # n_used=23, no neutrals
4 "Q2_FeatDiscovery": [5]*7 + [4]*16, # n_used=23, no neutrals
5 "Q3_AITitlesTags": [5]*12 + [4]*10 + [3]*1, # n_used=22, 1 neutral excluded
6 "Q4_MultiTopicTags": [5]*10 + [4]*11 + [3]*2, # n_used=21, 2 neutrals excluded
7 "Q5_ExplorerViews": [5]*9 + [4]*14, # n_used=23, no neutrals
8 "Q6_CalendarFilters": [5]*8 + [4]*15, # n_used=23, no neutrals
9 "Q7_BaselineComp": [5]*6 + [4]*14 + [3]*3} # n_used=20, 3 neutrals excluded
10 NEUTRAL = 3
11 print(f"{'Feature':<22} {'n_used':>6} {'W+':>8} {'p':>12} {'Reject H0':>10}")
12 print("-" * 62)
13 for name, r in ratings.items():
14     filtered = [x for x in r if x != NEUTRAL]
15     diffs = [x - NEUTRAL for x in filtered]
16     W, p = wilcoxon(diffs, alternative="greater")
17     print(f"{'name':<22} {'len(filtered)':>6} {'W':>8.1f} {'p':>12.6f} {'str(p<0.05):>10}")
18

```

Feature	n_used	W+	p	Reject H0
Q1_OverallDesign	23	276.0	0.000008	True
Q2_FeatDiscovery	23	276.0	0.000006	True
Q3_AITitlesTags	22	253.0	0.000011	True
Q4_MultiTopicTags	21	231.0	0.000018	True
Q5_ExplorerViews	23	276.0	0.000007	True
Q6_CalendarFilters	23	276.0	0.000006	True
Q7_BaselineComp	20	210.0	0.000021	True

For all seven questions $W^- = 0$, yielding $p < 0.001$ in every case. The n used varies because the Wilcoxon procedure excludes tied differences (neutral ratings produce zero difference from $H_0 = 3$). Q1,Q2,Q5,Q6 is notable in having no neutral responses excluded at all, producing $W^+ = n(n+1)/2 = 276$, the maximum possible sum of ranks at $n = 23$ and representing the strongest possible test result at this sample size.

14.3 Qualitative Results

Open-ended responses from Q8 ($n = 23$) were analyzed using thematic coding across three areas: organization, retrieval, and AI-supported search.

Organization: Several participants highlighted the topic-switch notification as an interesting and distinctive feature that helps maintain clearer conversation boundaries. The multi-topic tagging system was also viewed positively because it helps users understand conversations that include multiple themes. A small number of participants suggested making Pinned and Favorites sections collapsible, indicating possible refinements to the sidebar organization.

Retrieval: Participants noted that date-based grouping (Today, Yesterday, Older) made it easier to locate recent conversations. The grid layout in Chat Explorer was also mentioned as helpful because it allows users to quickly scan conversation titles, tags, and previews in one view. A few responses suggested that additional color differentiation between tags could further improve visual scanning.

Search and AI Features: Many participants highlighted the usefulness of AI-generated titles and tags for organizing conversations automatically. Participants noted that AI-generated categorization can reduce the effort required to manually organize chat histories. The ability to edit AI-generated titles and tags was also mentioned as important for maintaining user control and trust in the system.

13.4 Design Implications

The final evaluation results indicate that participants responded positively to the proposed interface features, with 87–100% positive responses across all survey questions. While a small number of neutral responses appeared for Q7 (comparison to existing interfaces), overall ratings still indicate that participants perceived the redesigned interface as an improvement for organizing and retrieving past conversations.

Qualitative feedback suggests that features such as topic-switch detection, AI-generated titles and tags, and structured browsing through Chat Explorer were particularly well received. At the same time, participants proposed small improvements such as color differentiation for tags, which could further improve usability in future iterations. These findings suggest that the overall design direction is promising while still leaving opportunities for incremental refinement.

REFERENCES

Figma Inc. (2024). *Figma: Collaborative interface design tool.* <https://www.figma.com>

Nielsen, J. (1994). Usability inspection methods. In Conference companion on Human factors in computing systems (pp. 413–414). ACM.

Norman, D. A. (2013). The design of everyday things (Revised ed.). Basic Books.

APPENDICES

Needfinding Survey Link :

<http://peersurvey.cc.gatech.edu/platform/survey-questions.html?id=267e6ad8d38044a39a57948b1b16fc26>

Survey: Organizing and Finding Past AI Conversations

1. How often do you use AI chat tools (ChatGPT, Claude, etc.)?

- Very Frequently
- Frequently
- Occasionally
- Rarely
- Never

2. About how full does your chat history feel to you?

- I only have a few chats
- I have some chats but can still recognize most of them
- I have many chats and need to scroll to find older ones
- I have so many chats that it feels cluttered

3. Do you regularly return to old chats?

- Very Frequently
- Frequently
- Occasionally
- Rarely
- Never

4. How organized does your current chat list feel?

- Very organized
- Somewhat organized
- Neutral
- Cluttered
- Very cluttered

5. When searching for an old chat, what is most frustrating?

- Too many results
- Hard to identify the correct one
- Results look similar
- Cannot filter by time or topic
- Projects or folders are hard to navigate

6. Do you find it easy to distinguish today's chats from older ones?

- Very easy
- Somewhat easy
- Difficult
- Very difficult

7. Do you currently use folders or projects to organize chats?

- Yes often

- Sometimes
- Rarely
- I do not use them

8. If you use folders or projects, are they easy to browse?

- Very easy
- Somewhat easy
- Difficult
- Very difficult
- I do not use them

9. How helpful would it be if chats were clearly separated into Today, Yesterday, and Older sections?

- Strongly Agree | Agree | Neutral | Disagree | Strongly Disagree

10. How helpful would a dedicated section for favorite chats be?

- Strongly Agree | Agree | Neutral | Disagree | Strongly Disagree

11. How helpful would a dedicated section for pinned chats be?

- Strongly Agree | Agree | Neutral | Disagree | Strongly Disagree

12. How helpful would it be to see all your projects displayed in a structured layout (similar to desktop folders with a list view), making it easier to browse and recognize them?

- Strongly Agree | Agree | Neutral | Disagree | Strongly Disagree

13. How helpful would it be to access recent project chats directly from the main chat list without navigating into each project?

- Strongly Agree | Agree | Neutral | Disagree | Strongly Disagree

14. How helpful would it be to view your chat history in a folder-style list view (instead of a single long scrolling list)?

- Strongly Agree | Agree | Neutral | Disagree | Strongly Disagree

15. Can you briefly describe a recent time you tried to find something in an old AI chat? What made it easy or difficult?

- [Short Answer Text]

RAW RESPONSES:

1. The title is often vague so I had to open the chat to know whether it contains what I'm looking for. It made it difficult to find what i was looking for. I usually end up using the search chat feature instead.

2. When I use AI for personal projects, it's typically Claude. And when I use Claude, all of my previous chats are displayed on the left hand side and usually have written titles. However, those written titles can be hard to differentiate.

3. I do not login frequently, so answered to the best of my thinking capability. I am generally able to scroll and find an old chat. But if the list is too long, scrolling wont be easy and I can see the need of a better search mechanism

4. It's difficult especially if the AI names the chat the first thing you talk about, and then the discussion evolves. It can be hard to pinpoint what the conversation contents were just from the title. Also just the sheer amount of chats to filter through makes it easier just to start over.

5. n/a

6. It was pretty difficult. I was using AI to help me understand the concepts in a lecture. I kept adding prompts in the same conversation, and if I want to find a specific topic, it was a bit difficult.

7. I forgot the exact wording of a previous chat so it was difficult to search for

8. Keyword search does not always return results in a chat I know contains the keyword. I have to manually check whether the keywords are actually in the chat the screen cannot load.

9. One time I was trying to look for AI generated lecture notes in my chats. It was difficult because the lectures are all very similar to each other, making it difficult to find the AI chat with a specific lecture.

10. At work I often set up chats with a lot of content with several code files before asking my question or doing the task. We are required to use internal AI tools that don't have these nice organizational features. So sometimes when interrupting a task it is difficult to go back to find the chat with all the context because it's a wall of random text
11. The title not representing all of its contents. The title reflected only the initial prompts.
12. Couldn't find old chats because I had to scroll so far, and a lot of the message were really long.
13. having to scroll through the chat history, it's time consuming
14. I often make multiple chats about a similar subject but different sub-tasks or sub-themes, and I get confused which is about what
15. I was looking for a workout plan I made last month. The chats had similar titles, so it took a while to find the right one.
16. I always use the search bar, and it works for me so far.
17. I don't organize them, so I had to manually look for it so it was difficult
18. I have multiple chat threads for same subject and if am trying to find a particular conversation, it becomes difficult to locate it
19. Easy is keyword search
20. the names of chatGPT chats don't correspond to their content, but rather the first question or two, making them hard to identify
21. I tried to find my old chat history, but the chat history was missing or wasn't visible for me which one was it. Another time is when I find the chat, but the section of the chat is too hard to find because the chat is too cluttered and AI gave lots of answers that I didn't need
22. I tried to find a conversation about course recommendations. It was easy to know when I got to the right chat since the topic was unique enough, but it was hard to find the chat mixed in with the other chats.

23. Big chats get pretty slow so I usually delete them. An individual chat is just a text history so I'd have to rely on a text search. If I don't name old chats descriptively enough they can be hard to find

24. I tend to do a bunch of unrelated queries in one single chat. Can be hard to find relevant info when other things are mentioned. end up wuth thimngs like: The "Tattoo" of Professionalism: Just like that cat spay tattoo, first impressions are permanent. Even though it's "casual" and "unscripted," wear a nice shirt and ensure your background is tidy.

25. I want to reuse a prompt template that I use months ago. It was difficult to find the conversation history as I need to guess the target query keyword multiple times before I get it right.

INITIAL EVALUATION SURVEY LINK:

<http://peersurvey.cc.gatech.edu/gt/c5d74d4fc92940cf9503e5495dcd693f>

GOOGLE SURVEY LINK:

<https://forms.gle/3i2Z8GS5PGxrGdrz8>

The following are the verbatim open-ended responses collected from 20 participants in response to the question: "What is one feature you found most helpful, and one thing that could be improved or felt unclear?"

1. Can improve Search functionality with context than specific keywords or tags.
2. Calendar Retrieval → Most helpful; Dynamic Content Suggestion → Not so clear.
3. The feature where the chats are organized as cards is really good. Also searching chats with dates is the most helpful. Overall a really robust UI. Everything for me felt clear.
4. Good Calendar Based Retrieval of Chats.
5. Timeline grouping is my favourite feature which eases my search. Felt Calendar based retrieval most helpful.
6. List/Grid View with Tags and Preview — shows chat name, preview, tags, and date for easy recognition.
7. Title and tag suggestions reduce the effort to organize the conversations.
8. To search older chats, I'd like it if the calendar can select a range of dates rather than just a single date. Also, I'd like search to include keywords that are matched to words in actual conversations.
9. Most helpful — Tags. Improvement: Colored tags.
10. Most helpful → Calendar Retrieval; Can be improved → Dynamic Content Suggestions.

11. Calendar based Retrieval is most helpful to track past data with accuracy and not spend time scrolling endlessly.
12. One feature that was helpful was the title feature and something that can be improved is the tag suggestion. I liked the 2nd, 3rd and 4th prototype concepts. Particularly the filtering and keyword search. Would be wary of cluttering the interface too much.
13. I really liked being able to see past chats in the grid view with keywords grouped with them. The calendar view is nice, but I think I would have trouble remembering the day if it was long enough ago.
14. Calendar is most helpful.
15. Grouping of chats into yesterday, today would be helpful.
16. I liked the first one the most because it was straightforward without being overwhelming and cluttering.
17. Love the grouping of chats into today, yesterday, and older, as well as the tagging improvements like multi-topic chats. Personally, I don't see the calendar-based search capability providing much value for my use cases; I can't picture myself knowing what day a chat took place if it was more than 2 or 3 days in the past.
18. Organizing historical chats and being able to pin. I thought the 'tags' method was a little unclear but likely learnable.
19. Tags.
20. The first one having the breakdown of the different chats.

Q1: Timeline Grouping (Organization)

Participants were shown a sidebar where chats are grouped into Today, Yesterday, and Older, along with options like Pin and Favorites.

Question asked:

"The grouping of chats into Today, Yesterday, and Older is clear and easy to understand."

Responses:

- Strongly Agree: 9/20 (45%)
- Agree: 7/20 (35%)
- Neutral: 3/20 (15%)
- Disagree: 1/20 (5%)

- Strongly Disagree: 0/20 (0%)

Interpretation:

A total of 80% of participants responded positively, indicating that timeline grouping is intuitive and helps users quickly understand chat organization.

Q2: Pin and Favorite Actions

Participants were shown hover-based pin and favorite interactions within the sidebar.

Question asked:

"The Pin and Favorite actions are easy to discover and understand."

Responses:

- Strongly Agree: 7/20 (35%)
- Agree: 8/20 (40%)
- Neutral: 3/20 (15%)
- Disagree: 2/20 (10%)
- Strongly Disagree: 0/20 (0%)

Interpretation:

While 75% of participants responded positively, some users found these actions less discoverable, suggesting a need for improved visibility.

Q3: List View with Tags and Preview (Chat Explorer)

Participants were shown a structured list view displaying title, preview, tags, and date.

Question asked:

"The list view provides enough information to understand what a conversation contains without opening it."

Responses:

- Strongly Agree: 10/20 (50%)
- Agree: 7/20 (35%)
- Neutral: 3/20 (15%)
- Disagree: 0/20 (0%)
- Strongly Disagree: 0/20 (0%)

Interpretation:

With 85% positive responses, this feature was highly effective in supporting quick recognition of conversations.

Q4: Calendar-Based Retrieval

Participants were shown a calendar interface used to retrieve chats by date.

Question asked:

"The calendar and project folder would help me retrieve older or related conversations quickly."

Responses:

- Strongly Agree: 8/20 (40%)
- Agree: 7/20 (35%)
- Neutral: 3/20 (15%)
- Disagree: 2/20 (10%)
- Strongly Disagree: 0/20 (0%)

Interpretation:

Although 75% responded positively, some users were unsure about relying on dates, indicating potential usability limitations.

Q5: Multi-topic Titles and Tags

Participants were shown grid view with AI-generated titles and a “multi-topic” tag.

Question asked:

“The interface clearly communicates when a chat contains multiple topics.”

Responses:

- Strongly Agree: 7/20 (35%)
- Agree: 8/20 (40%)
- Neutral: 3/20 (15%)
- Disagree: 2/20 (10%)
- Strongly Disagree: 0/20 (0%)

Interpretation:

Most users (75% positive) understood the concept, but some found it slightly unclear, suggesting a need for better explanation or visual cues.

Q6: Keyword and Tag-Based Search

Participants were shown a search interface with keyword highlights and visible tags.

Question asked:

“The search screen clearly shows why each result was returned.”

Responses:

- Strongly Agree: 11/20 (55%)
- Agree: 6/20 (30%)
- Neutral: 3/20 (15%)
- Disagree: 0/20 (0%)

- Strongly Disagree: 0/20 (0%)

Interpretation:

With 85% positive responses, search was one of the most successful features, especially due to transparency in results.

Q7: AI-Generated Title and Tag Suggestions

Participants were shown AI-generated title/tag suggestions within chats.

Question asked:

"AI-generated title and tag suggestions would reduce the effort required to organize conversations."

Responses:

- Strongly Agree: 9/20 (45%)
- Agree: 7/20 (35%)
- Neutral: 3/20 (15%)
- Disagree: 1/20 (5%)
- Strongly Disagree: 0/20 (0%)

Interpretation:

A strong 80% positive response indicates that users see value in automation for organization.

FINAL EVALUATION SURVEY

Peer Survey Link:

<http://peersurvey.cc.gatech.edu/gt/4c81a245cd8d4c5ea3ab202b00ddf3c0>

Google Survey Link:

https://docs.google.com/forms/d/e/1FAIpQLSc_wOZ5TxLyReEdVpYaxUC6MkOfKrwadN45EXtu7Deo5Qcufg/viewform?usp=publish-editor

Q1: The overall design was easy to understand when I first opened it. [Likert 1–5]

Q2: It was easy to find and understand the main features (pin, favorite, archive, search, calendar). [Likert 1–5]

Q3: The AI-generated titles and tags helped me quickly understand conversations. [Likert 1–5]

Q4: The Multi-Topic tag clearly indicates when a conversation contains multiple contexts or themes. [Likert 1–5]

Q5: The Chat Explorer views (list and grid) make it easy to browse and identify conversations. [Likert 1–5]

Q6: The calendar and date range filters in Chat Explorer make it easy to find older conversations. [Likert 1–5]

Q7: Compared to current ChatGPT or Claude interfaces, this design improves how I organize and retrieve past conversations. [Likert 1–5]

Q8: Do you have any feedback? Something you liked, confusing, or could be improved. [Open text]

Appendix F: Final Evaluation Open-Ended Responses (n=23, verbatim)

- If we can edit the colour theme of the chat, it will be nice.
- Liked almost all the new features added to ease users search. Kudos to the great work!
- The part where it detects off-topic conversations and asks us if we want to start a new chat looks like an interesting standout feature.
- I like the multi tagging for each chat feature and the simple deconstructed view of all the menus. Grid layout offers all important points of note in a single tile.
- Very well created.
- A good one overall. Felt better than ones I am already using.
- Innovative and Creative Development.
- Great product overall! I really liked how switching context within a chat would prompt to begin a new chat keeping overall focus consistent.
- The colors chosen can be improved in the UI, otherwise good overall.
- Overall good interface.
- Tags should be different colors.
- I really liked how it uses AI to make the different categories for better search and organization. If I would improve something maybe if there was a way to hide the pinned favorite and archives tab.
- No comments. Well done.
- I like the grouping of chats by dates (today, yesterday...).
- Date function is well-intended, but it might be better to prioritise the favourites/pinned chats.

Appendix H: Raw Brainstorming Ideas (24 Ideas Generated)

The following ideas were generated during the brainstorming session before convergence:

- Dynamic AI-generated chat titles that update as conversations evolve

- Multi-topic compound titles for conversations that shift subject mid-session
- Multi-topic badge/indicator on chat rows in list and grid views
- User-defined project folders mirroring desktop file structure
- Drag-and-drop conversation assignment to project folders
- Pinning frequently accessed chats to top of sidebar
- Favoriting chats for a dedicated instant-access sidebar section
- Timeline grouping: Today / Yesterday / Older / This Month
- Calendar widget for date-based retrieval in sidebar
- Date-range selection (click-and-drag) on calendar widget
- Quick-filter shortcuts: Last 7 Days, Last 30 Days, This Month
- Keyword highlighting in yellow within search result previews
- Semantic tag-based retrieval (surface chats by tag proximity, not only exact keyword)
- Auto-generated tags from conversation content via AI
- User-editable AI tag suggestions (accept, modify, replace)
- Visual distinction between system-generated and user-defined tags
- Grid view of chats showing title, preview, tags, date per card
- List view with sortable columns (date, name, tag count)
- Topic-switch detection popup during active conversations
- Onboarding tooltip sequence for low-discoverability features
- Archived conversations section to declutter main list
- Recycle bin for soft-deleted conversations with restore option
- Color-coded tags by topic category for improved scanability
- Progressive disclosure of advanced filter options to reduce default view complexity